

Re: the $SO(3)$, $l = 1$ warm-up for the three-sphere harmonics problem — how the “honest” matrices were built

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This note only addresses the $SO(3)$ / $l = 1$ harmonic sign-transpose question raised in your PDF. It is a warm-up calculation on S^2 , isolating the operator/generator convention issue before we move to the full S^3 spinor-harmonic construction — it does not yet say anything about that larger question.

1 The two comparisons, side by side

Your own check computes, for a candidate abstract matrix T'_i and the combined column of basis functions $v = (\psi_{1,1}; \psi_{1,0}; \psi_{1,-1})$, the ordinary matrix-vector product $T'_i \cdot v$, and compares it entrywise to $(\hat{L}_i \psi_{1,1}; \hat{L}_i \psi_{1,0}; \hat{L}_i \psi_{1,-1})$.

That is a well-defined thing to compute, but it is a *different* operation from asking: “what matrix, in the basis $\{\psi_{1,1}, \psi_{1,0}, \psi_{1,-1}\}$, actually represents $\hat{L}_i$?”

The second question is answered like this: apply $\hat{L}_i$ to *one basis function at a time*, and expand the result back in the same basis:

$$\hat{L}_i \psi_{1,m} = \sum_{m'} M_{m'm} \psi_{1,m'},$$

so that column m of the matrix M is exactly the (unique) set of coefficients needed to write $\hat{L}_i \psi_{1,m}$ back in the basis. This M is the honest matrix representation of $\hat{L}_i$.

If $(T'_i \cdot v)_{\text{row}}$ happens to equal $\hat{L}_i \psi_{1,\text{row}}$ for every row, that tells you T'_i and M are related by a **transpose**: $(T'_i \cdot v)_{\text{row}} = \sum_{\text{col}} (T'_i)_{\text{row,col}} \psi_{1,\text{col}}$ uses the *rows* of T'_i , whereas the honest construction of M above reads off the *columns*. For a diagonal or otherwise symmetric matrix the row and column pictures coincide, which is exactly why your $\hat{L}_z = T'_z$ check gave a clean match — $T'_z = \text{diag}(1, 0, -1)$ is trivially symmetric. It is not a coincidence that the one clean component was the diagonal one.

2 Building M three separate computational routes

I built the honest matrix M_x (and M_y , M_z) three separate ways, to make sure the conclusion did not depend on which technique was used — these are three computational cross-checks rather than three fully independent derivations, since all three use the same operators, basis functions, and conventions throughout.

Method A — direct solve. Apply $\hat{L}_x$ to each of $\psi_{1,1} = \sin \theta e^{i\phi}$, $\psi_{1,0} = \cos \theta$, $\psi_{1,-1} = \sin \theta e^{-i\phi}$ in turn, then solve the linear system for the coefficients that re-express each result in the same basis (matching the $e^{i\phi}$, constant, and $e^{-i\phi}$ parts separately).

Method B — coefficient extraction. Same idea, but the coefficients are extracted by evaluating the residual at a few sample values of ϕ and solving directly, rather than a general symbolic solve — a cheap cross-check on Method A.

Method C — inner-product (bra-ket) projection. The cleanest of the three, and the one I would recommend as the standard going forward:

$$M_{m'm} = \frac{\langle \psi_{1,m'} | \hat{L}_i | \psi_{1,m} \rangle}{\langle \psi_{1,m'} | \psi_{1,m'} \rangle}, \quad \langle f | g \rangle := \int_0^\pi \int_0^{2\pi} \bar{f} g \sin \theta \, d\phi \, d\theta.$$

This is exactly the ordinary quantum-mechanical way of reading off a matrix element, and it sidesteps the row/column ambiguity entirely, since each entry of M is defined by its own independent integral. The simple form above (dividing by $\langle \psi_{1,m'} | \psi_{1,m'} \rangle$ alone) relies on $\{\psi_{1,1}, \psi_{1,0}, \psi_{1,-1}\}$ being an orthogonal set, which they are here; for a non-orthogonal basis one would need the full Gram-matrix inversion, $M = G^{-1}A$ with $A_{m'm} = \langle \psi_{m'} | \hat{L}_i | \psi_m \rangle$ and $G_{m'm} = \langle \psi_{m'} | \psi_m \rangle$.

All three methods gave, for your specific choice $\chi_{1,1} = \chi_{1,-1} = \sin \theta$, $\chi_{1,0} = \cos \theta$, exactly:

$$M_x = \begin{pmatrix} 0 & -\frac{1}{2} & 0 \\ -1 & 0 & 1 \\ 0 & \frac{1}{2} & 0 \end{pmatrix}, \quad M_y = \begin{pmatrix} 0 & \frac{i}{2} & 0 \\ -i & 0 & -i \\ 0 & \frac{i}{2} & 0 \end{pmatrix}, \quad M_z = \text{diag}(1, 0, -1).$$

Comparing these directly to your own T'_x, T'_y, T'_z (equations 19–21 of your note):

$$M_x = -(T'_x)^\top, \quad M_y = +(T'_y)^\top, \quad M_z = T'_z,$$

each an exact identity (zero residual), independent of which of the three methods above produced M .

3 Why y and z looked clean and x did not — the real reason

A clarification on terminology first: M above is the matrix acting on *coefficient* columns (if $\Psi = (\psi_{1,1}, \psi_{1,0}, \psi_{1,-1})$ is treated as the basis itself and a state is a coefficient vector c in that basis, M sends $c \mapsto Mc$). Your own comparison instead treats Ψ as the column being acted on directly, so it tests $M^\top$, not M — consistent with Section 1.

I initially described the match for y, z as “coincidental,” which was sloppy and not quite right. Since $\psi_{1,1}, \psi_{1,0}, \psi_{1,-1}$ are linearly independent, an exact match across all three rows is a genuine matrix identity, not a numerical accident tied to one test vector. The real explanation is more structural, and traces back to your own S matrix directly.

Identify $\psi_{1,1} = x + iy$, $\psi_{1,0} = z$, $\psi_{1,-1} = x - iy$ (exactly, with $x = \sin \theta \cos \phi$, $y = \sin \theta \sin \phi$, $z = \cos \theta$ — no extra normalization needed). Applying $\hat{L}_i$ to the plain Cartesian triple (x, y, z) gives, for *all three* generators at once:

$$\hat{L}_i(x, y, z)^\top = -T_i(x, y, z)^\top,$$

using your *original* (un-diagonalized) generators T_x, T_y, T_z from equations (14)–(16). This is a single, uniform sign — with the active/passive convention used in your equations, a differential operator realizing a rotation of *functions* carries the opposite sign to the same rotation acting on the coordinates themselves. It is not itself the source of the asymmetry between x and y, z .

The asymmetry comes from matching your S -diagonalized ordering to the $(\psi_{1,1}, \psi_{1,0}, \psi_{1,-1})$ ordering you actually compare against. Writing $X = (x, y, z)^\top$, direct computation gives $SX = (-\psi_{1,-1}, \psi_{1,0}, -\psi_{1,1})^\top$ — the *same* three functions as $\Psi = (\psi_{1,1}, \psi_{1,0}, \psi_{1,-1})^\top$, but reordered and with two sign flips. That reordering is exactly the matrix

$$C = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad C^2 = I, \quad \Psi = CSX.$$

Combining the uniform sign above with this reordering gives, for every generator at once,

$$\hat{L}_i \Psi = -C T'_i C^{-1} \Psi,$$

which I have checked directly matches $\hat{L}_i \Psi$ exactly for $i = x, y, z$. The point is what C -conjugation does to each T'_i :

$$C T'_x C^{-1} = T'_x \quad (\text{unchanged}), \quad C T'_y C^{-1} = -T'_y, \quad C T'_z C^{-1} = -T'_z.$$

Combined with the uniform “ -1 ” from the active/passive sign, x picks up $(-1)(+1) = -1$ relative to T'_x , while y, z pick up $(-1)(-1) = +1$ relative to T'_y, T'_z — exactly reproducing $M_x = -(T'_x)^\top$, $M_y = +(T'_y)^\top$, $M_z = T'_z$. The x -generator is singled out because C (forced by the specific ordering and sign convention you used for $\psi_{1,\pm 1}$) happens to commute with T'_x but anti-commute with T'_y and T'_z — a systematic consequence of the basis reordering, not an accident of specific numerical entries.

In short: your arithmetic in equations (1)–(29) is correct throughout — I rechecked every one of them independently and found no error — and the -1 on x alone has this precise, traceable origin, not an arbitrary one.

4 One small, unrelated thing, for completeness

Separately: in your equation (18), S^{-1} has a sign slip in one entry (the $(2, 3)$ position). It did not affect anything downstream — your own T'_x, T'_y, T'_z in equations (19)–(21) are exactly what you get from the *correct* inverse, so nothing needs to be revisited there.

5 Looking forward to your reasoning

I’d be very interested to see why you were expecting $\hat{L}_i \psi = T'_i \psi$ to hold identically for all three generators — if there’s a different, perhaps more physically natural convention where that holds exactly (rather than with the same basis ordering, phase choices, and action convention used above), that would be worth understanding properly before we go further with the three-sphere case.